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Senator Mike Weissman
Legislative Audit Committee
State Capital Building
200 E. Colfax Ave. Room 346
Denver, CO 80203

Re: Colorado's Risk-Limiting Election Audit

Dear Senator Weissman:

I invented Risk-Limiting Audits (RLAs) in 2007. I helped draft Colorado's RLA legislation and helped Colorado pilot and implement RLAs, including working as a consultant to the Colorado Secretary of State on a grant from the US Election Assistance Commission. The CORLA software is based on my original implementations of RLAs. More recently, I worked with Boulder County on RLAs of ranked-choice voting using newer software. A brief summary of my work in election integrity appears at the end of this letter.

In 2012, David Wagner and I introduced Evidence-Based Elections, a paradigm for conducting elections that ensures elections produce convincing public evidence that reported election outcomes are correct.

Colorado deserves praise for being the first state to conduct routine, statewide RLAs. Colorado's RLAs presumably give election officials considerable comfort that the audited contests were decided correctly. But changes are needed for Colorado's elections and audits to justify public confidence that election outcomes are correct.

The guiding principle of Colorado's elections and audits should be to ensure that the public has access to enough information to assess whether election outcomes are trustworthy. For the RLA, that means the public needs to be able to confirm that the audit did not stop before it should have. The details the public must observe depend on details of the RLA method. Colorado already publishes or broadcasts some important details, such as the software and the selection of the "seed," but more information is needed–see below.

RLA methods have progressed substantially since the CORLA software was written. To the best of my knowledge, the only substantive changes to the CORLA software in the last decade have been to support auditing IRV/RCV contests, a change made as an add-on rather than re-architecting the software to use the latest and most efficient approaches. Newer sampling and auditing methods make it practical to audit every contest in every election. They unify audits of a vast range of social choice functions, including plurality, multi-winner plurality, supermajority, instant-runoff voting (IRV/RCV), STAR voting, approval voting, and more.

Here are a number of suggestions for improving Colorado's RLAs.

More trustworthy inputs. I understand that Colorado's RLA relies on the voting system for the total number of ballots and ballot cards–crucial inputs to RLAs. That reliance assumes that every validly cast card was scanned and tabulated by the system exactly once and that the software correctly reported the total number of cards. But the whole point of auditing is that the voting system can misbehave and that procedural errors may occur. Indeed, some of the largest errors I've encountered in vote tallies resulted from failing to scan some batches of ballot cards or scanning the same batch more than once–errors that would not be detected without a physical inventory of the ballot cards separate from the voting system. It is far better to use physical inventories, SCORE, and other administrative records to get upper bounds on the number of cards cast in each contest without relying on the voting system. Colorado should also confirm that its RLA software properly accounts for unaccounted-for ballot cards (cards that are not in the ballot manifest); to the best of my knowledge, CORLA currently does not.

Contest selection. Auditing some contests provides no evidence that any other contest was called correctly, so every contest should receive scrutiny. Colorado should audit more contests to a risk limit, including–if not especially–contests with small margins. One strength of Colorado's RLA practice is that auditors collect data "opportunistically" on every contest that appears on any audited ballot card, not just the "targeted" contests, and thereby provides some evidence about whether those other contests were called correctly. Colorado should report the "measured risk" for every contest, that is, the risk limit for which an RLA of the contest would have terminated for the sample actually drawn. Workloads for auditing small contests and contests with small margins can be vastly reduced using “style-based sampling,” which makes it feasible to audit every contest in an election to a risk limit. See below. The contests to audit should not be selected at the discretion of the Secretary of State or any other entity–the selection itself can be politicized or could be used to hide problems. Nor should contests be selected on the basis of expected workload, or small-margin contests will never be audited, even though they are most susceptible to outcome-changing errors or problems.

I understand that the Secretary of State generally directs counties to audit one statewide contest and one county-level contest. I understand that in lieu of auditing a county-level contest, the SOS sometimes directs counties to audit their “portion” of a cross-jurisdictional contest. That practice does not yield an RLA: no audit can provide evidence that a cross-jurisdictional contest was called correctly by sampling from only one jurisdiction in which votes were cast. In my opinion, it is largely a waste of time. The practice should be eliminated.

Voter privacy. Colorado redacts some cast vote records (CVRs) to protect voter privacy. I understand that currently CVRs are redacted “by row” (i.e., suppressing entire CVRs) rather than “by column” (eliminating some contests from some CVRs). It would be preferable to redact by column, which would preserve more of the data for public scrutiny.

Audit transparency. Ensuring that the cards that were randomly selected are indeed the cards that were retrieved and inspected is key to the validity of an RLA. The mapping from random numbers to selected ballots should be published to allow the public to verify that the correct ballots were selected. The public also needs to be able to determine whether the correct cards were retrieved and inspected. This can be done in a variety of ways; it might help to have a trusted, independent observer involved in the process. The public should be able to see each audited card and its corresponding CVR. If CVRs are redacted, risk calculations should substitute the “least favorable” possible vote for any redacted vote.

Additional integrity safeguards. It is crucial that the audit have access to every validly cast ballot card. The trustworthiness of the collection of cast ballots is key. To ensure that only eligible voters cast votes, eligibility audits should be extended beyond signature verification, which is known to be error-prone. To ensure that the paper trail has not been compromised, compliance audits should be extended.

Technical improvements to CORLA software. As mentioned previously, the science and engineering of RLAs has advanced considerably in the 15 or so years since CORLA was written. Colorado should move from sampling all ballot cards using equal probabilities to style-based sampling, which generally reduces sample sizes and makes it practical to audit small contests. CORLA software uses risk-measuring functions that are unnecessarily conservative. Replacing the current risk calculations with the SHANGRLA framework and the ALPHA risk-measuring function will improve efficiency (reduce sample sizes) and allow all social choice functions (including IRV) to be audited using the same approach and core functions.

Sincerely,

Philip B. Stark

Bio

I am Distinguished Professor of Statistics at the University of California, Berkeley, where I have served as department chair and associate dean. In 2007 I invented "risk-limiting audits" ("RLAs"), endorsed by the National Academies of Science, Engineering, and Medicine and the American Statistical Association, among others, and required or authorized by law in about 15 states. I published the first academic paper about RLAs and helped conduct the first RLA pilots in 2008. I designed and helped conduct the first dozen pilot RLAs in California and Colorado, helped draft RLA legislation for several states, and I've published open-source software to support RLAs. In 2012, David Wagner and I introduced "evidence-based elections," a paradigm for conducting demonstrably trustworthy elections. I have served on the Board of Advisors of the US Election Assistance Commission and its cybersecurity subcommittee, the Board of Directors of Verified Voting Foundation, the Board of Directors of the Election Integrity Foundation, the Board of Advisors of the Open Source Election Technology Institute, the Board of Advisors of Scrutineers, and on the California Post Election Audit Standards Working Group. I have worked with the Secretaries of State of California, Colorado, and New Hampshire and numerous local election officials. I have testified about election integrity in state and federal courts and to legislators. I received the IEEE Cybersecurity Award for Practice, the UC Berkeley Chancellor's Award for Research in the Public Interest, and the John Gideon Award for Election Integrity. I am a fellow of the American Statistical Association and the Institute of Physics and a member of the American Academy of Arts and Sciences.

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